

1. Verify if master is in clean state.

```
Nandu@Nandu MINGW64 ~/GitDemo (master)
$ git status
On branch master
Your branch is up to date with 'origin/master'.

nothing to commit, working tree clean
```

2. Create a branch “GitWork”. Add a file “hello.xml”.

```
Nandu@Nandu MINGW64 ~/GitDemo (master)
$ git branch Gitwork

Nandu@Nandu MINGW64 ~/GitDemo (master)
$ git branch
  Gitwork
* master
```

```
Nandu@Nandu MINGW64 ~/GitDemo (master)
$ git checkout Gitwork
Switched to branch 'GitWork'

Nandu@Nandu MINGW64 ~/GitDemo (GitWork)
$ git branch
  Gitwork
  master

Nandu@Nandu MINGW64 ~/GitDemo (GitWork)
$ notepad++ hello.xml
```

3. Update the content of “hello.xml” and observe the status

```
Nandu@Nandu MINGW64 ~/GitDemo (GitWork)
$ git status
On branch GitWork
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    hello.xml

nothing added to commit but untracked files present (use "git add" to track)
```

4. Commit the changes to reflect in the branch

```
Nandu@Nandu MINGW64 ~/GitDemo (GitWork)
$ git add hello.xml

Nandu@Nandu MINGW64 ~/GitDemo (GitWork)
$ git commit -m "Added hello.xml in Gitwork"
[Gitwork c407a14] Added hello.xml in Gitwork
1 file changed, 1 insertion(+)
create mode 100644 hello.xml
```

5. Switch to master.

6. Add a file “hello.xml” to the master and add some different content than previous.

```
MINGW64:/c/Users/Nandu/GitDemo
Nandu@Nandu MINGW64 ~/GitDemo (GitWork)
$ git checkout master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.

Nandu@Nandu MINGW64 ~/GitDemo (master)
$ notepad++ hello.xml

Nandu@Nandu MINGW64 ~/GitDemo (master)
$ git status
On branch master
Your branch is up to date with 'origin/master'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    hello.xml

nothing added to commit but untracked files present (use "git add" to track)
```

7. Commit the changes to the master

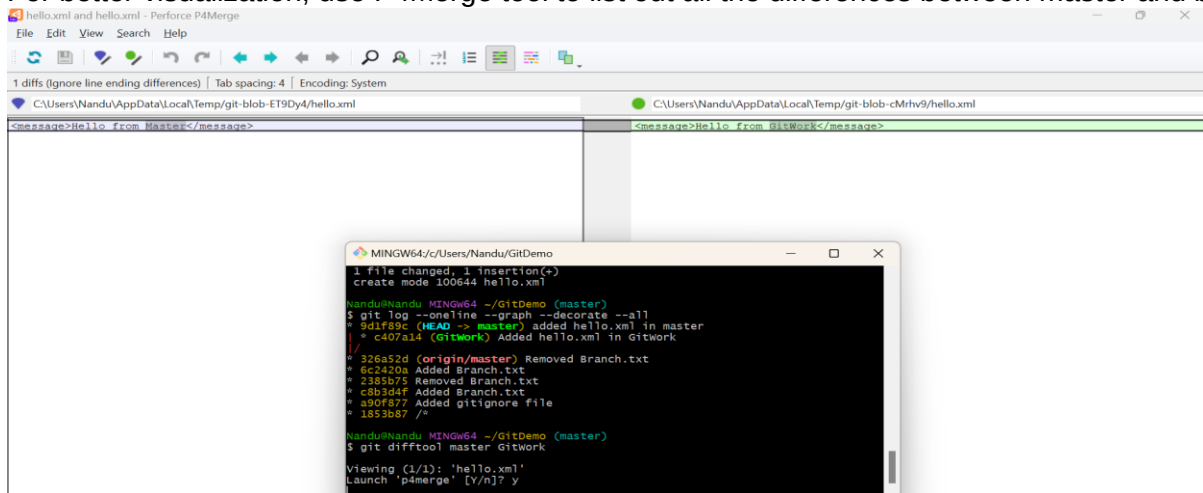
```
Nandu@Nandu MINGW64 ~/GitDemo (master)
$ git commit -m "added hello.xml in master"
[master 9d1f89c] added hello.xml in master
1 file changed, 1 insertion(+)
create mode 100644 hello.xml
```

8. Observe the log by executing “git log –oneline –graph –decorate –all”

```
Nandu@Nandu MINGW64 ~/GitDemo (master)
$ git log --oneline --graph --decorate --all
* 9d1f89c (HEAD -> master) added hello.xml in master
| * c407a14 (GitWork) Added hello.xml in GitWork
|/
* 326a52d (origin/master) Removed Branch.txt
* 6c2420a Added Branch.txt
* 2385b75 Removed Branch.txt
* c8b3d4f Added Branch.txt
* a90f877 Added gitignore file
* 1853b87 /*
```

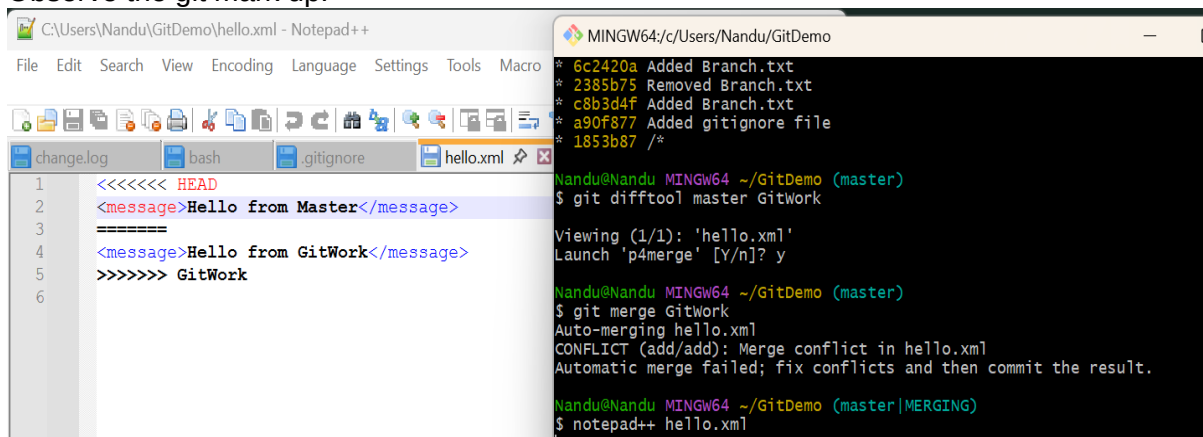
9. Check the differences with Git diff tool

10. For better visualization, use P4Merge tool to list out all the differences between master and branch

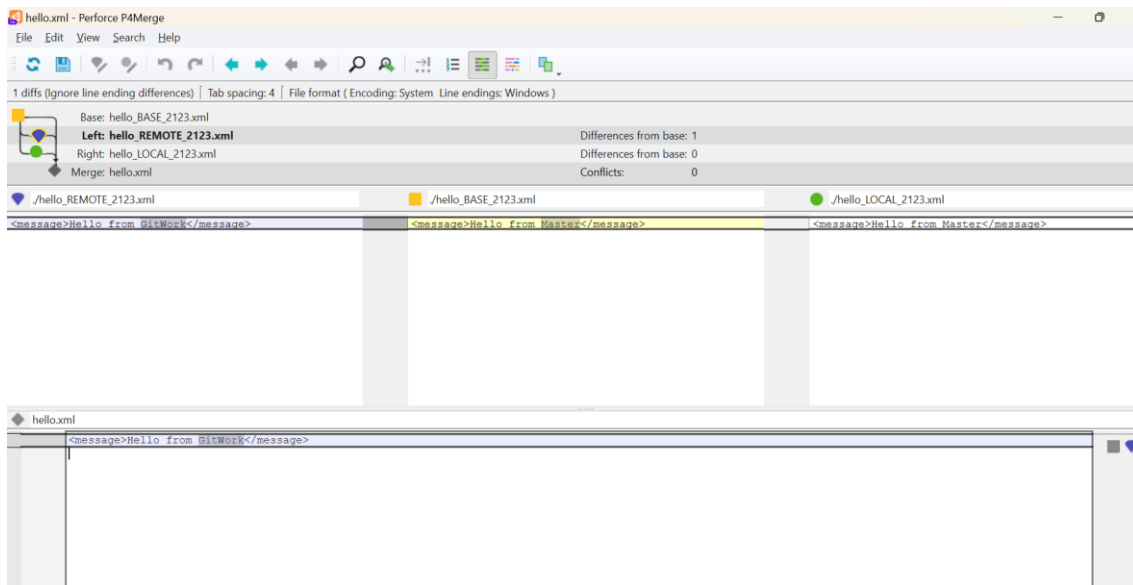


11. Merge the bran to the master

12. Observe the git mark up.



13. Use 3-way merge tool to resolve the conflict



14. Commit the changes to the master, once done with conflict
15. Observe the git status and add backup file to the .gitignore file.

```
Nandu@Nandu MINGW64 ~/GitDemo (master|MERGING)
$ git commit
[master 0e979cd] Merge branch 'GitWork'

Nandu@Nandu MINGW64 ~/GitDemo (master)
$ git status
On branch master
Your branch is ahead of 'origin/master' by 3 commits.
  (use "git push" to publish your local commits)

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    hello.xml.orig

nothing added to commit but untracked files present (use "git add" to track)
```

16. Commit the changes to the .gitignore
17. List out all the available branches

```
Nandu@Nandu MINGW64 ~/GitDemo (master)
$ git branch
  GitWork
* master
```

18. Delete the branch, which merge to master.

```
Nandu@Nandu MINGW64 ~/GitDemo (master)
$ git branch -d GitWork
Deleted branch GitWork (was c407a14).
```

19. Observe the log by executing “git log --oneline --graph --decorate”

```
Nandu@Nandu MINGW64 ~/GitDemo (master)
$ git log --oneline --graph --decorate
* 23534c7 (HEAD -> master) Added .orig files to gitignore
* 0e979cd Merge branch 'GitWork'
|
| * c407a14 Added hello.xml in GitWork
* | 9d1f89c added hello.xml in master
|/
* 326a52d (origin/master) Removed Branch.txt
* 6c2420a Added Branch.txt
* 2385b75 Removed Branch.txt
* c8b3d4f Added Branch.txt
* a90f877 Added gitignore file
* 1853b87 /*
```